Smartphone Bedtime Use and Sleep Quality: A Survey of University Students in Malaysia

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Abstract

Smartphone use before sleep is common among university students, yet little is known about how it relates to sleep quality in Southeast Asia. We surveyed 612 undergraduates (N = 612) across three Malaysian universities. Students who used their phone for more than an hour after lights-out reported 41% more nights of poor sleep than those who did not. Bedtime use was associated with later sleep onset (r = .32, p < .001). These findings suggest that campus sleep campaigns should target late-night phone habits.

Keywords: sleep quality, smartphone use, university students, Malaysia

1. Introduction

Poor sleep is widespread among university students and is linked to lower academic performance and wellbeing (Hershner & Chervin, 2014). Smartphones have become the last thing many students look at before sleeping. However, little is known about how bedtime smartphone use relates to sleep quality among students in Southeast Asia, where most research has not been conducted. Prior work from Europe and North America suggests that screen light and engaging content delay sleep onset (Exelmans & Van den Bulck, 2016). The aim of this study was to examine whether the amount of smartphone use after lights-out predicts sleep quality and sleep onset in Malaysian undergraduates.

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2. Method

We conducted a cross-sectional online survey in March 2025. The questionnaire measured bedtime smartphone use, sleep quality using the Pittsburgh Sleep Quality Index, and sleep onset time.

2.1 Participants

A total of 612 undergraduates aged 18 to 25 years (M = 20.8, SD = 1.6) were recruited from three public universities. Most participants were female (64%).

2.2 Measures and analysis

Bedtime use was measured as self-reported minutes of phone use after lights-out on a typical night. We used multiple regression to test whether bedtime use predicted sleep quality, controlling for age, gender and caffeine intake.

3. Results

Students who used their phone for more than an hour after lights-out reported 41% more nights of poor sleep than students with no bedtime use. Bedtime use was positively correlated with later sleep onset (r = .32, p < .001). In the regression model, bedtime use remained a significant predictor of poorer sleep quality (β = .27, p < .001) after controlling for covariates. Students who kept their phone outside the bedroom were 2.1 times more likely to report good sleep quality. Table 1 shows descriptive statistics for all variables.

4. Discussion

Our findings suggest that bedtime smartphone use is an important and modifiable factor in student sleep. The association between late-night use and later sleep onset is consistent with studies from Western samples (Exelmans & Van den Bulck, 2016). These results highlight the need

for universities to include phone habits in sleep and wellbeing programmes. Simple strategies such as keeping phones outside the bedroom may help students sleep better.

5. Limitations

This study has several limitations. The cross-sectional design prevents conclusions about causality. Sleep and phone use were self-reported, which may introduce recall bias. Future research should use objective measures such as screen-time logs and actigraphy.

6. Conclusion

Bedtime smartphone use is common and is linked to poorer sleep among Malaysian university students. Reducing phone use after lights-out is a practical target for campus health campaigns.

References

Exelmans, L., & Van den Bulck, J. (2016). Bedtime mobile phone use and sleep in adults. Social Science & Medicine, 148, 93-101.

Hershner, S. D., & Chervin, R. D. (2014). Causes and consequences of sleepiness among college students. Nature and Science of Sleep, 6, 73-84.